

Zomato Business Intelligence Project

Business Insights Report | Orders Jan 2023 - Dec 2024

Executive Summary

Ten insights from 20,475 orders, 9,254 customers and 1,200 restaurants. Each has evidence from the analytical tables, the business implication, and a recommended action. The overall picture: demand is healthy and evenly spread, but about a third of order value is lost to failed orders, the causes are not explained by the data currently captured, and several features (membership, coupons, ratings) show no measurable effect.

#	Insight	Priority*
1	A third of order value never becomes revenue	High
2	Failures are not caused by peak-hour load	High
3	Demand is concentrated at lunch and dinner	Medium
4	Late deliveries vary by city more than by traffic or vehicle	High
5	Delivery time cannot be predicted with the current data	High
6	Cancellation rates differ by cuisine and city	Medium
7	Membership tiers show no behavioural difference	Medium
8	Most customers order once or twice	High
9	Coupons cost Rs 0.76M with no measurable effect on failures	Medium
10	Ratings do not reflect service outcomes	Medium

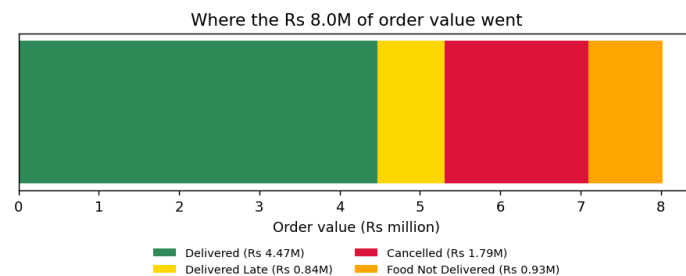
**Priority is the analyst's judgement, based on the size of the value at stake and how actionable the finding is. It is not a calculated score.*

How to read this report

Revenue means FinalAmount of orders with status Delivered or Delivered Late. Late % is measured against delivered orders. Rupee estimates are simple what-if calculations and assume affected orders would have had the same average value; they are not forecasts. Where a difference is small or based on small samples, the report says so.

Insight 1: A third of order value never becomes revenue

Insight. Of Rs 8.02M in order value placed, Rs 2.71M (33.8%) was on orders that were cancelled or where the food was never delivered.



Evidence

- Cancelled: 22.3% of orders, Rs 1.79M.
- Food not delivered: 11.5% of orders, Rs 0.93M.
- Only 55.6% of orders are delivered on time; another 10.5% are delivered late.

Business implication

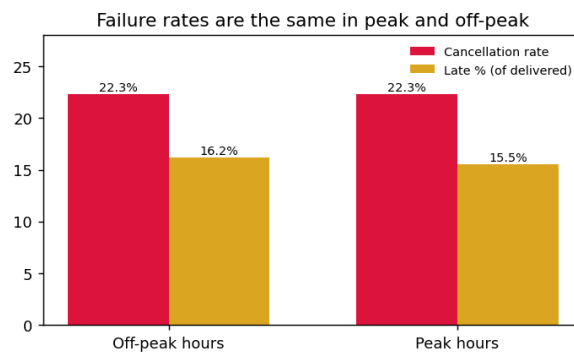
Order failures are the largest single lever on revenue in this dataset - larger than any pricing, cuisine or city effect we can measure. Each percentage point of cancellation avoided is worth about Rs 80k over the two years (what-if estimate).

Recommended action

Set a target for cancellation and non-delivery rates, report them weekly, and start capturing a cancellation reason and who cancelled (customer, restaurant or platform). The data has no reason field today, which is the main blocker to diagnosis.

Insight 2: Failures are not caused by peak-hour load

Insight. Cancellation and late rates are essentially the same in peak and off-peak hours, so lunch and dinner congestion does not explain why orders fail.



Evidence

- Cancellation rate: 22.3% off-peak vs 22.3% peak.
- Late rate: 16.2% off-peak vs 15.5% peak.
- The same holds for weekday vs weekend and for customers in the same city as the restaurant.

Business implication

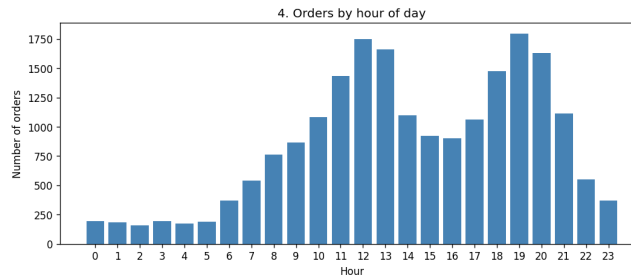
Adding delivery partners at peak times may help other goals, but it is unlikely to fix the failure rate on its own. The causes look systemic (process, restaurant acceptance, dispatch) rather than load-related.

Recommended action

Investigate restaurant acceptance and preparation, payment failures and partner assignment rules before investing in peak-hour capacity.

Insight 3: Demand is concentrated at lunch and dinner

Insight. 44.2% of orders fall inside the two meal-time windows (12-2 pm and 7-9 pm); the four busiest hours (12, 13, 19, 20) alone carry 33.4%.



Evidence

- Order volume peaks at 12-1 pm and 7-8 pm (chart below).
- Order volume per day is the same on weekdays and weekends (about 26 orders per day each).
- Revenue is flat across months (Rs 5.31M total), so timing within the day is the main pattern.

Business implication

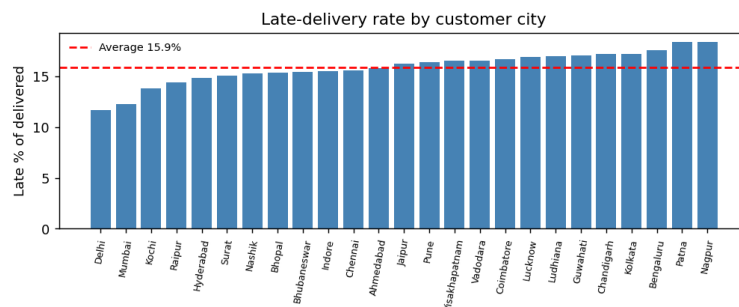
Predictable daily peaks make scheduling and promotions easy to target. Combined with Insight 2, they are better used for planning partner shifts and off-peak offers than for explaining failures.

Recommended action

Schedule partner shifts around the four peak hours and test off-peak promotions (mid-afternoon, late evening) to smooth demand.

Insight 4: Late deliveries vary by city more than by traffic or vehicle

Insight. 15.9% of delivered orders are late, and late orders take about 64 minutes versus 34 for on-time ones. City explains more of the variation than traffic, vehicle or partner rating.



Evidence

- Late rate by city ranges from 11.7% (Delhi) to 18.4% (Nagpur).
- Traffic effect is mild: 14.3% in low traffic to 18.0% in severe traffic.
- Vehicle type: 14.2% (Bicycle) to 16.4% (Bike). Partner rating shows no relationship (about 15-16% in every rating band).
- City differences are based on roughly 500 delivered orders per city, so extreme values may partly be chance.

Business implication

Lateness is a local operations question. A network-wide fix is less likely to help than a look at the highest-late cities.

Recommended action

Review dispatch and partner supply in the highest-late cities (Nagpur, Patna, Bengaluru) and compare practices with the best performers (Delhi, Mumbai). Re-check after another quarter of data.

Insight 5: Delivery time cannot be predicted with the current data

Insight. Four regression models were trained on 15 pre-delivery features. None explained delivery time (R-squared of about 0).

Evidence

- Best models: Linear Regression MAE 11.64 min, R-squared 0.001; Random Forest MAE 11.66 min, R-squared -0.003.
- Delivery time has a standard deviation of about 15 minutes; simply predicting the average gives an error of about 11.5 minutes, the same as the models.
- Correlation of delivery time with traffic, rain, basket size, ratings and weather is 0.03 or lower.
- Distance, the most obvious driver, is unavailable: customers have no coordinates and only 4% of orders have customer and restaurant in the same city.

Business implication

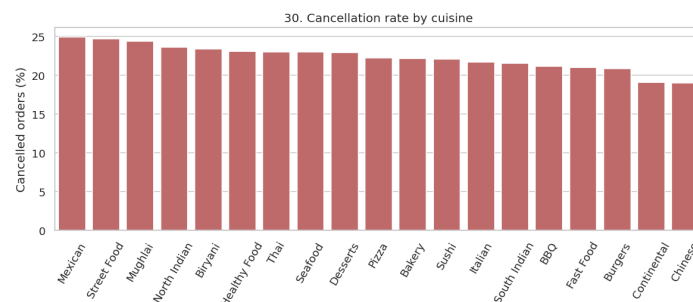
A customer-facing delivery-time estimate built on these features would be unreliable and should not be deployed.

Recommended action

Capture customer location, restaurant handover time and partner GPS. Retrain once distance and preparation time are available. The pipeline (model_delivery.py) is ready to rerun.

Insight 6: Cancellation rates differ by cuisine and city

Insight. Cancellations run from about 19% to 25% depending on cuisine and from 20% to 27% depending on city. The gaps are modest but worth monitoring.



Evidence

- Lowest cuisines: Chinese (19.0%), Continental (19.1%). Highest: Mexican (24.9%), Street Food (24.6%).
- Lowest cities: Indore (20.0%), Delhi (20.7%). Highest: Kolkata (26.6%), Patna (24.4%).
- Payment method (21-24%) and membership (22-23%) show no meaningful differences.

Business implication

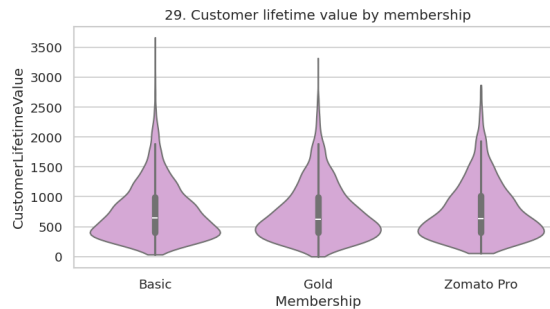
Cuisine and city are better starting points for investigation than payment method or membership. With about 800-1,100 orders per group, some of the spread may be random variation.

Recommended action

Track the top-cancelling cuisines and cities monthly and confirm the gap persists before acting. Pair this with the cancellation-reason data from Insight 1.

Insight 7: Membership tiers show no behavioural difference

Insight. Basic, Gold and Zomato Pro customers spend, order and cancel at the same levels.



Evidence

- Average order value: Rs 392 (Basic), Rs 391 (Gold), Rs 390 (Pro).
- Average customer lifetime value: Rs 751, Rs 747, Rs 760.
- Cancellation rate is 22-23% in every tier.

Business implication

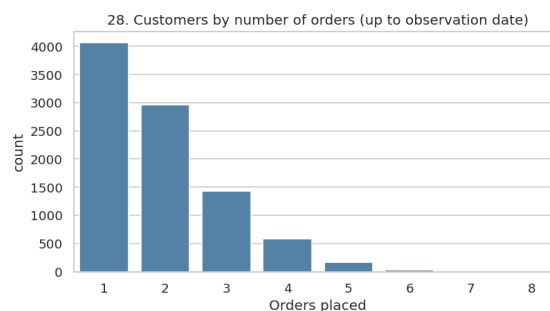
Whatever benefits memberships provide are not visibly changing how much or how often customers order, so the paid tiers may not be earning their cost. This dataset cannot show whether membership affects retention over a longer period.

Recommended action

Review membership benefits and pricing, and run a small test (for example a benefit tied to order frequency) to see whether behaviour changes.

Insight 8: Most customers order once or twice

Insight. Customers average 1.9 orders, and 43.9% have ordered only once. Past behaviour does not predict who orders again.



Evidence

- 75.9% of customers have at most two orders (up to the observation date).

- Churn (no order in the next 60 days) is about 88% in every recency and order-count group.
- Four churn models reached ROC-AUC of about 0.50, no better than guessing; an 'everyone churns' rule scores higher on accuracy and F1.

Business implication

Retention is the growth problem, but this data cannot identify at-risk customers. The practical lever is converting first-time customers into second-time customers.

Recommended action

Trial a second-order incentive shortly after a first delivery, and collect behavioural signals (app opens, complaints, refunds, delivery issues) so a churn model has something to learn from.

Insight 9: Coupons cost Rs 0.76M with no measurable effect on failures

Insight. 35.2% of orders use a coupon, and total discounts equal 9.7% of food cost. The data does not show a measurable benefit.

Evidence

- Total discount value: Rs 0.76M.
- Cancellation rate: 21.9% with a coupon vs 22.5% without.
- Coupon orders have lower final value, partly because the discount is subtracted from FinalAmount, so this cannot be read as customers ordering less.

Business implication

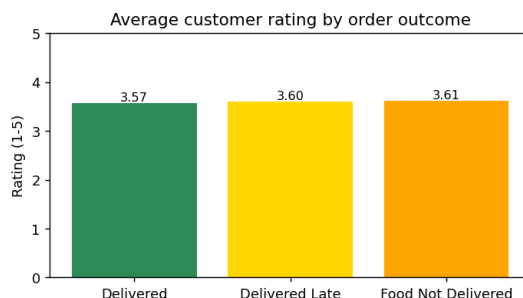
There is no evidence here that coupons reduce cancellations, and the dataset cannot show whether they create extra orders. Discounts are a real cost either way.

Recommended action

Run a holdout test (a group of comparable customers receiving no coupon) to measure incremental orders before increasing coupon spend, and cap discount rate per order in the meantime.

Insight 10: Ratings do not reflect service outcomes

Insight. Customer ratings are almost identical whether the order was on time, late, or never delivered, so ratings cannot currently be used as a service-quality measure.



Evidence

- Average rating: 3.57 (Delivered), 3.60 (Delivered Late), 3.61 (Food Not Delivered).
- 87.2% of 'Food Not Delivered' orders still have a customer rating.
- Restaurant rating has near-zero correlation with revenue (-0.05) and cancellation rate (0.02).

Business implication

Either the feedback is not linked to the order outcome or ratings are captured differently than assumed. Metrics built on ratings, including 40% of PerformanceScore, may not measure quality.

Recommended action

Check how feedback is collected and linked to orders. Until that is confirmed, treat the Rating weight in PerformanceScore as a design choice to revisit, and rely more on cancellation and late rates for restaurant performance.

Caveats and Next Steps

- This is an observational dataset; the report describes patterns, not causes.
- About 46% of orders lack weather and traffic readings, and 1,173 orders have no valid date (excluded from time-based views).
- The ML models reach roughly chance level, which is a finding about the data rather than the method. Adding distance, preparation time and behavioural signals is the main way to improve them.
- Rupee figures are what-if estimates that assume affected orders would have had the average value.

Suggested order of work: (1) capture cancellation reasons and customer location, (2) investigate failures by restaurant and city, (3) run coupon and second-order experiments, (4) revisit membership and rating design.